

# MATH2030J - Discrete Mathematics

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## Lecture 5



上海交通大學

SHANGHAI JIAO TONG UNIVERSITY

Prepare for Thursday's discussion session.

- ▶ Come see me during the break if you haven't met your coursework teammates yet.
- ▶ By Wednesday evening: upload your solutions as a pdf to the appropriate Canvas assignment.

# Tentative Schedule

## Week 1: Introduction and Logic

**Handed out:** DS1 (discussion sheet 1) on Sun1

## Week 2: Logic

**Due:** ► DS1 exercises by Wed2 evening.

**Handed out:** HW1 on Wed2

**RC:** time and location TBD

**TA OH:** time and location TBD

## Week 3: Sets, Introduction to Algorithms

**Due:** • HW1 on Thu3, 23:59

**Handed out:** DS2 on Wed3

**RC:** time and location TBD

**TA OH:** time and location TBD

Propositional Logic (0th-order logic):

- ▶ Compound propositions
- ▶ Truth tables for logical operators
- ▶ Conditional statements
- ▶ Tautological equivalence
- ▶ Satisfiability

Rules of Inference:

- ▶ Definitions: argument, premises, conclusion, validity of an argument.
- ▶ Argument form, validity of an argument form.
- ▶ Rules of inference are valid argument forms.
- ▶ Formal implication.

- ▶ Truth trees / semantic tableaux.

Predicate logic (1st-order logic):

- ▶ Predicate = propositional function of propositional variable(s).
- ▶ Quick reminder – sets: naive definition, construction, inclusion, set operators.
- ▶ Quantifiers:  $\forall, \exists, \exists!$ . Definitions, scope, rules of manipulation.

### 1.3. Predicates and Quantifiers

- ▶ Nested quantifiers
- ▶ Axiomatic set theory examples
- ▶ Trees for predicate logic

### 1.4. Introduction to Formal Proofs

- ▶ Terminology

- ▶ Axiomatic set theory
- ▶ Set-theoretic proofs
- ▶ Truth trees for first order logic

## 1.4. Introduction to Formal Proofs

- ▶ Simple axiomatic systems
- ▶ Natural deduction systems
- ▶ Inference rules: introduction and elimination
- ▶ Example proofs
- ▶ Curry-Howard correspondence
- ▶ Limits of naive set theory

## 1.5. (Informal) Proof Techniques

- ▶ Enumeration and examples

# Ch. 1. Logic and Proofs

- ▶ Propositional Logic
- ▶ Rules of Inference
- ▶ Predicates and Quantifiers
- ▶ Introduction to Formal Proofs
- ▶ (Informal) Proof Techniques



- ▶ Runze Cai, MATH1160J Sp. 2024 course
- ▶ K. H. Rosen, Discrete Mathematics and its Applications
- ▶ J. Gallier, Discrete Mathematics
- ▶ Frank Pfenning, Constructive Logic course:  
<https://www.cs.cmu.edu/~fp/courses/15317-f09/schedule.html>

## 1.4. Introduction to Formal Proofs

# Formal and Informal Proofs

- ▶ A *proof* is some kind of *deduction (derivation)* that proceeds from a set of *hypotheses (premises, axioms)* in order to derive a *conclusion*, using some *logical rules*.
- ▶ Proofs can be very informal, using a set of loosely defined logical rules, possibly omitting steps and premises.
- ▶ Proofs can be completely formal, using a very clearly defined set of rules and premises. Such proofs are usually processed or produced by programs called proof checkers and theorem provers (LEAN, COQ, etc.).
- ▶ Thus, a human prover evolves in a spectrum of formality:
  - ▶ It is extremely time- (and space-)consuming to write or read fully formal proofs: it is rarely the best choice.
  - ▶ But there are critical systems (e.g. flight-control, nuclear reactor, ...) in which we must have absolute confidence.

# Axiomatic Set Theory Examples

- ▶ Extensionality Axiom

$$\forall A, B (\forall x (x \in A \leftrightarrow x \in B) \rightarrow A = B)$$

- ▶ Empty Set Axiom

$$\exists B \forall x (x \notin B)$$

- ▶ Pairing Axiom

$$\forall u, v \exists B \forall x (x \in B \leftrightarrow x = u \vee x = v)$$

- ▶ Union Axiom

$$\forall a, b \exists B \forall x (x \in B \leftrightarrow x \in a \vee x \in b)$$

- ▶ Powerset Axiom

$$\forall a \exists b \forall x (x \in b \leftrightarrow x \subseteq a)$$

where  $x \subseteq a$  is shorthand for  $\forall t (t \in x \rightarrow t \in a)$ .

## Basic Facts

$$x \in A \cup B \Leftrightarrow (x \in A) \vee (x \in B)$$

$$x \notin A \cup B \Leftrightarrow (x \notin A) \wedge (x \notin B)$$

$$x \in A \cap B \Leftrightarrow (x \in A) \wedge (x \in B)$$

$$x \notin A \cap B \Leftrightarrow (x \notin A) \vee (x \notin B)$$

$$x \in A - B \Leftrightarrow (x \in A) \wedge (x \notin B)$$

$$x \notin A - B \Leftrightarrow (x \notin A) \vee (x \in B)$$

$$A \subseteq B \Leftrightarrow \forall x, (x \in A \rightarrow x \in B)$$

## Basic Facts

$$x \in A \cup B \Leftrightarrow (x \in A) \vee (x \in B)$$

$$x \notin A \cup B \Leftrightarrow (x \notin A) \wedge (x \notin B)$$

$$x \in A \cap B \Leftrightarrow (x \in A) \wedge (x \in B)$$

$$x \notin A \cap B \Leftrightarrow (x \notin A) \vee (x \notin B)$$

$$x \in A - B \Leftrightarrow (x \in A) \wedge (x \notin B)$$

$$x \notin A - B \Leftrightarrow (x \notin A) \vee (x \in B)$$

$$A \subseteq B \Leftrightarrow \forall x, (x \in A \rightarrow x \in B)$$

## Remark

- ▶ Prove something exists: sufficient to find an example.
- ▶ Prove  $P(x)$  for all  $x \in A$ : Take any  $x \in A$  and continue. (or use induction if  $A = \mathbb{N}$ , more on this later.)

## Example

Given sets  $A, B, C$ , if  $A \subseteq B$  and  $B \cap C = \emptyset$ , then  $A \cap C = \emptyset$ .

## Proof.

Note that in first-order logic, the statement is

$$\forall x(x \in A \rightarrow x \in B), \forall x \neg(x \in B \wedge x \in C) \vdash \forall x \neg(x \in A \wedge x \in C)$$

Let  $x$  be an arbitrary element, it suffices to show the following proposition

$$x \in A \rightarrow x \in B, \neg(x \in B \wedge x \in C) \vdash \neg(x \in A \wedge x \in C)$$

Let  $p, q, r$  be the logical variables for  $x \in A, x \in B, x \in C$ , resp., we need to show

$$p \rightarrow q, \neg(q \wedge r) \vdash \neg(p \wedge r)$$

(which is clearly true since it is equivalent to  $p \rightarrow q, p \wedge r \vdash q \wedge r$ )

## Proof.

We can use truth table.

| $p$ | $q$ | $r$ | $p \rightarrow q$ | $\neg(q \wedge r)$ | $\neg(p \wedge r)$ | $[(p \rightarrow q) \wedge \neg(q \wedge r)] \rightarrow \neg(p \wedge r)$ |
|-----|-----|-----|-------------------|--------------------|--------------------|----------------------------------------------------------------------------|
| 0   | 0   | 0   | 1                 | 1                  | 1                  | 1                                                                          |
| 0   | 0   | 1   | 1                 | 1                  | 1                  | 1                                                                          |
| 0   | 1   | 0   | 1                 | 1                  | 1                  | 1                                                                          |
| 0   | 1   | 1   | 1                 | 0                  | 1                  |                                                                            |
| 1   | 0   | 0   | 0                 | 1                  | 1                  |                                                                            |
| 1   | 0   | 1   | 0                 | 1                  | 0                  |                                                                            |
| 1   | 1   | 0   | 1                 | 1                  | 1                  | 1                                                                          |
| 1   | 1   | 1   | 1                 | 0                  | 0                  |                                                                            |



# Set-Theoretic Proofs

Or use truth trees.

$$p \rightarrow q, \neg(q \wedge r) \vdash \neg(p \wedge r)$$

|    |                                                                             |                   |
|----|-----------------------------------------------------------------------------|-------------------|
| 1. | $p \rightarrow q \checkmark$                                                | premise           |
| 2. | $\neg(q \wedge r) \checkmark$                                               | premise           |
| 3. | $p \wedge r \checkmark$                                                     | $\neg$ consequent |
| 4. | $p$                                                                         | 3                 |
| 5. | $r$                                                                         | 3                 |
|    | <div><div><math>\swarrow</math></div><div><math>\searrow</math></div></div> |                   |
| 6. | $\neg p$<br><b>X</b><br>4, 6                                                | 1                 |
|    | <div><div><math>\swarrow</math></div><div><math>\searrow</math></div></div> |                   |
| 7. | $\neg q$<br><b>X</b><br>6, 7                                                | 2                 |
|    | <div><div><math>\swarrow</math></div><div><math>\searrow</math></div></div> |                   |
|    | $\neg r$<br><b>X</b><br>5, 7                                                |                   |

[https://www.umsu.de/trees/#p~5q,~3\(q~1r\)%7C=~3\(p~1r\)](https://www.umsu.de/trees/#p~5q,~3(q~1r)%7C=~3(p~1r))

# Set-Theoretic Proofs

Or use truth trees for the original sequent in first-order logic.

( $Ax := x \in A$  etc.)

$$\forall x(Ax \rightarrow Bx), \forall x\neg(Bx \wedge Cx) \vdash \forall x\neg(Ax \wedge Cx)$$

|                                                              |                                      |                   |
|--------------------------------------------------------------|--------------------------------------|-------------------|
| 1.                                                           | $\forall x(Ax \rightarrow Bx) *$     | premise           |
| 2.                                                           | $\forall x\neg(Bx \wedge Cx) *$      | premise           |
| 3.                                                           | $\exists x(Ax \wedge Cx) \checkmark$ | $\neg$ consequent |
| 4.                                                           | $At \wedge Ct \checkmark$            | 3                 |
| 5.                                                           | $At$                                 | 4                 |
| 6.                                                           | $Ct$                                 | 4                 |
| 7.                                                           | $\neg(Bt \wedge Ct) \checkmark$      | 2                 |
| 8.                                                           | $At \rightarrow Bt \checkmark$       | 1                 |
| <div style="text-align: center;">└───┬───<br/>└───┴───</div> |                                      |                   |
| 9.                                                           | $\neg At$<br><b>X</b><br>9, 5        | 8                 |
| <div style="text-align: center;">└───┬───<br/>└───┴───</div> |                                      |                   |
| 10.                                                          | $\neg Bt$<br><b>X</b><br>10, 9       |                   |
|                                                              | $\neg Ct$<br><b>X</b><br>10, 6       |                   |

# Truth Trees for First-Order Logic

$\exists y \forall x Pxy \vdash \forall x \exists y Pxy$

|    |                                  |                   |
|----|----------------------------------|-------------------|
| 1. | $\exists y \forall x Pxy$ ✓      | premise           |
| 2. | $\exists x \forall y \neg Pxy$ ✓ | $\neg$ consequent |
| 3. | $\forall y \neg Pay$ *           | 2                 |
| 4. | $\forall x Pxb$ *                | 1                 |
| 5. | $\neg Pab$                       | 3                 |
| 6. | $Pab$                            | 4                 |
|    | <b>X</b>                         |                   |
|    | 5, 6                             |                   |

<https://www.umsu.de/trees/#~7y~6xPxy%7C=~6x~7yPxy>

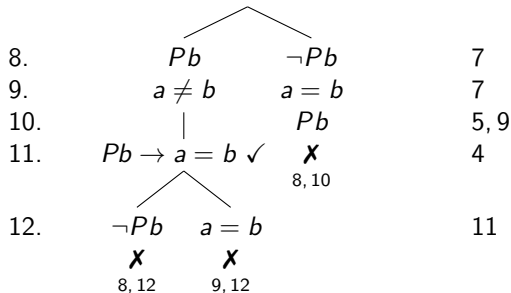
## Strategies

- ▶ Push negation past all quantifiers, but no further
- ▶ Introduce new variables for  $\exists$  (if applicable)
- ▶ Introduce existing variables for  $\forall$  (if applicable)
- ▶ Can reuse  $\forall$ -sentences

# Truth Trees for First-Order Logic

$\exists x(\forall y(Py \rightarrow x = y) \wedge Px) \vdash \exists x\forall y(Py \leftrightarrow x = y)$

- |    |                                                                   |                   |
|----|-------------------------------------------------------------------|-------------------|
| 1. | $\exists x(\forall y(Py \rightarrow x = y) \wedge Px) \checkmark$ | premise           |
| 2. | $\forall x\exists y\neg(Py \leftrightarrow x = y) *$              | $\neg$ consequent |
| 3. | $\forall y(Py \rightarrow a = y) \wedge Pa \checkmark$            | 1                 |
| 4. | $\forall y(Py \rightarrow a = y) *$                               | 3                 |
| 5. | $Pa$                                                              | 3                 |
| 6. | $\exists y\neg(Py \leftrightarrow a = y) \checkmark$              | 2                 |
| 7. | $\neg(Pb \leftrightarrow a = b) \checkmark$                       | 6                 |



# Simple Axiomatic System (Hilbert)

## Axioms

- ▶ (H1)  $p \rightarrow q \rightarrow p$
- ▶ (H2)  $(p \rightarrow q \rightarrow r) \rightarrow (p \rightarrow q) \rightarrow p \rightarrow r$
- ▶ (H3)  $(\neg p \rightarrow \neg q) \rightarrow q \rightarrow p$

## Inference Rules

Only modus ponens:

- ▶ (MP)  $p, p \rightarrow q \vdash q$

$\vdash A \rightarrow A$

1.  $A \rightarrow (A \rightarrow A) \rightarrow A$  (H1)
2.  $(A \rightarrow (A \rightarrow A) \rightarrow A) \rightarrow (A \rightarrow (A \rightarrow A)) \rightarrow (A \rightarrow A)$  (H2)
3.  $(A \rightarrow (A \rightarrow A)) \rightarrow (A \rightarrow A)$  1,2,(MP)
4.  $A \rightarrow A \rightarrow A$  (H1)
5.  $A \rightarrow A$  3,4,(MP)

# Natural Deduction System

## Axioms

None

## Inference Rules

- ▶ Negation introduction:  $p \rightarrow q, p \rightarrow \neg q \vdash \neg p$
- ▶ Negation elimination:  $\neg p \vdash p \rightarrow r$
- ▶ Double negation introduction:  $p \vdash \neg\neg p$
- ▶ Double negation elimination:  $\neg\neg p \vdash p$
- ▶ Conjunction introduction:  $p, q \vdash p \wedge q$
- ▶ Conjunction elimination:  $p \wedge q \vdash p$  ;  $p \wedge q \vdash q$
- ▶ Disjunction introduction:  $p \vdash p \vee q$  and  $q \vdash p \vee q$
- ▶ Disjunction elimination:  $p \vee q, p \rightarrow r, q \rightarrow r \vdash r$
- ▶ Modus ponens:  $p, p \rightarrow q \vdash q$
- ▶ Conditional proof:  $(p \vdash q) \vdash p \rightarrow q$

and others (for  $\leftrightarrow$ ,  $\exists$ ,  $\forall$ , etc.)

# Natural Deduction Rules (Classical Logic)

- ▶  $\wedge$ -introduction rule:

$$\frac{A \quad B}{A \wedge B} \wedge I \quad A, B \vdash A \wedge B$$

Here,  $A$  and  $B$  are **schematic variables**, and  $\wedge I$ , “conjunction introduction”, is the name of the **inference rule**. The  $\wedge$  is introduced in the **conclusion**:  $A \wedge B$ .

- ▶ General form of an inference rule:

$$\frac{J_1 \dots J_n}{J} \text{ name}$$

**Premises:**  $J_1, \dots, J_n$ ; **conclusion:**  $J$ .

# Natural Deduction Rules (Classical Logic)

- ▶  $\wedge$ -elimination rule (left / right conjunction elimination):

$$A, B \vdash A \qquad \frac{A \wedge B}{A} \wedge E_1 \qquad \frac{A \wedge B}{B} \wedge E_2 \qquad A, B \vdash B$$

- ▶  $\top$ -introduction:

$$\frac{}{\top} \top I$$

“Truth” should always be true: hence the introduction rule has no premises. Consequently, knowing  $\top$  gives us no information, so there is no elimination rule.

$\vdash A$  : “A is true”



## Hypothetical Judgements

Let  $A, B, C$  be arbitrary propositions. By right and left conjunction elimination:

$$\frac{\frac{A \wedge (B \wedge C)}{B \wedge C} \wedge E_2}{B} \wedge E_1$$

Have we proved that  $B$  is true?

# Hypothetical Judgements

Let  $A, B, C$  be arbitrary propositions. By right and left conjunction elimination:

$$\frac{A \wedge (B \wedge C)}{B \wedge C} \wedge E_2$$
$$\frac{B \wedge C}{B} \wedge E_1$$

$$A \wedge (B \wedge C) \vdash B$$

$$A, B, C \vdash B$$

Have we proved that  $B$  is true?

- ▶ All inferences are correct, but the first judgement, that  $A \wedge (B \wedge C)$  is true, has not been justified.
- ▶ We have proved that if  $A \wedge (B \wedge C)$  is true, then  $B$  is true: this is a hypothetical judgement.
- ▶ We have not proved that  $B$  is true:  $B$  is a schematic variable representing any proposition.

$$\vdash B$$

# Hypothetical Judgements

General form:

$$\begin{array}{c} J_1 \quad \dots \quad J_n \\ \vdots \\ J \end{array}$$

- ▶  $J_1, \dots, J_n$  are unproven assumptions, and  $J$  is the conclusion.
- ▶ We can substitute a proof for any hypothesis  $J_i$  (substitution principle for hypotheses).

Example:  $P \wedge Q \vdash Q \wedge P$

$$\frac{\frac{P \wedge Q}{Q} \wedge E_2 \quad \frac{P \wedge Q}{P} \wedge E_1}{Q \wedge P} \wedge I$$

$J_1 \quad \vdots \quad J_2$

- $\rightarrow$ -introduction (implication introduction) rule:

$[A]^a$

$\vdots$

$B$

$\frac{}{A \rightarrow B} \rightarrow I, a$

**Correct.** The tricky part of this rule is the label  $a$ . If we omit this annotation, the rule would read:

$\vdash (A \rightarrow B)$

- The tree on the left indicates that the conclusion “ $A \rightarrow B$  true” has been derived from the hypothesis “ $A$  true”. **Wrong.**

**WRONG:**

$A$

$\vdots$

$B$

$\frac{}{A \rightarrow B} \rightarrow I$

- The assumption  $A$  true is introduced in the process of proving  $A \rightarrow B$  true, so the conclusion should not depend on that assumption!
- We label  $a$  that assumption:  $[A]^a$ , as well as the inference rule that introduces it into the derivation:  $I, a$
- We say that the assumption “ $A$  true” labeled  $a$  is **discharged** in the last inference.
- It is critical that a discharged hypothesis is no longer available for reasoning, and that all labels introduced in a derivation are distinct.

Is each of the following proofs a hypothetical derivation or not?

Example 1:

$$P \wedge Q \vdash Q \wedge P$$

$$\frac{\frac{P \wedge Q}{Q} \wedge E_2 \quad \frac{P \wedge Q}{P} \wedge E_1}{Q \wedge P} \wedge I$$

Example 2:

$$\vdash A \rightarrow (B \rightarrow (A \wedge B))$$

$$\frac{\frac{[A]^a \quad [B]^b}{A \wedge B} \wedge I}{B \rightarrow (A \wedge B)} \rightarrow I, b$$

$$\frac{B \rightarrow (A \wedge B)}{A \rightarrow (B \rightarrow (A \wedge B))} \rightarrow I, a$$

$$A, B \vdash A \wedge B$$

$$\frac{A \quad B}{A \wedge B} \wedge I$$

$$A \vdash (B \rightarrow (A \wedge B))$$

$$\frac{A \quad [B]^b}{A \wedge B} \wedge I$$

$$\frac{A \wedge B}{B \rightarrow (A \wedge B)} \rightarrow I, b$$

Is each of the following proofs a hypothetical derivation or not?

### Example 1:

$$\frac{\frac{P \wedge Q}{Q} \wedge E_2 \quad \frac{P \wedge Q}{P} \wedge E_1}{Q \wedge P} \wedge I$$

### Example 2:

$$\frac{\frac{\frac{[A]^a \quad [B]^b}{A \wedge B} \wedge I}{B \rightarrow (A \wedge B)} \rightarrow I, b}{A \rightarrow (B \rightarrow (A \wedge B))} \rightarrow I, a$$

► Example 1 is a hypothetical derivation: it states that  $Q \wedge P$  is true if  $P \wedge Q$  is true, a hypothesis that has not been discharged. It proves:  $P \wedge Q \vdash Q \wedge P$ .

► Example 2 is not hypothetical, since all hypotheses have been discharged. It proves that “it is true, that  $A \rightarrow (B \rightarrow (A \wedge B))$ ”, which we can denote by

$$\vdash A \rightarrow (B \rightarrow (A \wedge B)).$$

- $\rightarrow$ -elimination (implication elimination) rule:

$$\frac{A \rightarrow B \quad A}{B} \rightarrow E$$

(MP : modus ponens)  
 $A \rightarrow B, A \vdash B$

- $\vee$ -introduction (disjunction introduction) rule:

$$A \vdash A \vee B$$

$$\frac{A}{A \vee B} \vee I_1$$

$$\frac{B}{A \vee B} \vee I_2$$

$$B \vdash A \vee B$$

- $\vee$ -elimination (disjunction elimination) rule:

$$\frac{A \vee B \quad \begin{array}{c} [A]^a \\ \vdots \\ C \end{array} \quad \begin{array}{c} [B]^a \\ \vdots \\ C \end{array}}{C} \vee E, a$$

$$\frac{A \vee B \quad \begin{array}{c} [A]^a \\ \vdots \\ C \end{array} \quad \begin{array}{c} [B]^a \\ \vdots \\ C \end{array}}{C} \vee E, a$$

$$A \vee B \vdash C$$

$[A]^a$  and  $[B]^a$  share a label because they are discharged in the same inference:  $\vee E, a$ .

- ▶  $\neg$ -introduction rule:

$$\frac{\begin{array}{c} [A]^a \\ \vdots \\ \perp \end{array}}{\neg A} \neg I, a$$

- ▶  $\neg$ -elimination rule:

$$\frac{\neg A \quad A}{\perp} \neg E$$

- ▶  $\perp$ -elimination rule:

$$\frac{\perp}{A} \perp E$$

- ▶ Proof-by-contradiction/indirect-proof rule:

$$\frac{\begin{array}{c} [\neg A]^a \\ \vdots \\ \perp \end{array}}{A} \text{by\_contra}, a$$



## Example Proofs

$$P \wedge Q \vdash Q \wedge P$$

$$\frac{\frac{P \wedge Q}{Q} \wedge E_2 \quad \frac{P \wedge Q}{P} \wedge E_1}{Q \wedge P} \wedge I$$

$$\frac{Q \quad P}{Q \wedge P} \wedge I$$

$$P \wedge Q \vdash P \wedge (P \wedge Q)$$

$$\frac{\frac{P \wedge Q}{P} \wedge E_1 \quad P \wedge Q}{P \wedge (P \wedge Q)} \wedge I$$

## Example Proofs

$P \wedge Q \vdash (P \wedge P) \wedge Q$

$$\frac{\frac{\overline{P} \quad \overline{P}}{P \wedge P} \wedge I \quad \overline{Q}}{(P \wedge P) \wedge Q} \wedge I$$

$$\frac{\frac{\frac{P \wedge Q}{P} \wedge E_1 \quad \frac{P \wedge Q}{P} \wedge E_1}{P \wedge P} \wedge I \quad \frac{\frac{P \wedge Q}{Q} \wedge E_2}{(P \wedge P) \wedge Q} \wedge I$$

$(P \rightarrow Q), (Q \rightarrow R) \vdash (P \rightarrow R)$

$$\begin{array}{c}
 [P]^a \\
 \vdots \\
 R \\
 \hline
 P \rightarrow R
 \end{array}
 \rightarrow I, a$$

$$\begin{array}{c}
 \begin{array}{c}
 P \rightarrow Q \quad [P]^a \\
 \hline
 Q \rightarrow R \quad Q \rightarrow E
 \end{array}
 \rightarrow E \\
 \hline
 R \\
 \hline
 P \rightarrow R
 \end{array}
 \rightarrow I, a$$

## Example Proofs

$P \rightarrow Q, Q \rightarrow R \vdash P \rightarrow R$

$$\frac{\frac{Q \rightarrow R}{\frac{R}{P \rightarrow R} \rightarrow I, a} \quad \frac{\frac{P \rightarrow Q \quad [P]^a}{Q} \rightarrow E}{\rightarrow E} \rightarrow E$$

$\vdash P \rightarrow Q \rightarrow P \wedge Q$

$$\frac{\frac{\frac{[P]^a \quad [Q]^b}{P \wedge Q} \wedge I}{Q \rightarrow P \wedge Q} \rightarrow I, b}{P \rightarrow Q \rightarrow P \wedge Q} \rightarrow I, a$$